# Tim Lichtenberg

Curriculum Vitae

Assistant Professor | Kapteyn Astronomical Institute | University of Groningen

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*I study the physical and chemical processes that drive the growth and long-term evolution of rocky planets and exoplanets, from formation through billions of years of geodynamic and climate evolution. My research traces how radioactive heating and planetesimal differentiation set the initial volatile budgets, how magma ocean epochs forge the first atmospheres, and how mantle dynamics and volatile cycling shape planetary environments over geological time. By bridging planetary physics, geochemistry, and exoplanet observations, my research group works to understand the conditions for the origins of life and its cosmic prevalence.*

## Research Positions

Tenure Track Assistant Professor & Branco Weiss Fellow (University of Groningen, NL) 10/2022–

Simons Collaboration on the Origins of Life Postdoctoral Fellow (University of Oxford, UK) 10/19–09/22

Swiss National Science Foundation Early.Postdoc Mobility Fellow (University of Oxford, UK) 08/18–09/19

## Education

Dr. sc. ETH Zurich (Department of Earth Sciences, ETH Zurich, Switzerland) 09/14–07/18

Thesis: Thermal Evolution of Forming Planets: Isotope Enrichment, Differentiation & Volatile Retention

Master of Science Physics (Georg-August-Universität Göttingen, Germany) 09/12–08/14

Thesis: Modeling gravitational instabilities in compact and massive protoplanetary disks with adaptive mesh refinement techniques (grade: 1.0)

Semester abroad (Peking University, People's Republic of China) 09/12–01/13

Studies of Chinese language and culture

Bachelor of Science Physics (Georg-August-Universität Göttingen, Germany) 09/09–08/12

Thesis: Constraining Exoplanet Characteristics with Asteroseismology (grade: 1.0)

Abitur 08, Wilhelmsgymnasium Kassel; military service 08–09, Schwarzenborn/Germany –09

## Honours & Awards

Europlanet Early-Career Medal, Europlanet Society 09/25

Planetary and Solar System Sciences Division Outstanding Early Career Scientist Award, European Geosciences Union (EGU) 04/23

Winton Award for Early Achievement in Geophysics, Royal Astronomical Society 01/22

Doctoral Thesis Award 2019, German Astronomical Society 09/19

PhD Prize for Planetary Systems and Astrobiology 2018, International Astronomical Union 06/19

European Astrobiology Network Association, Horneck-Brack Presentation Award, 2nd prize 09/18

Master's diploma awarded 'with distinction', Georg-August-Universität Göttingen 08/14

'Dr. Berliner - Dr. Ungewitter' award for M.Sc. thesis, Georg-August-Universität Göttingen 08/14

## Grants & Fellowships

*17 grants and fellowships in total, incl. 9 as PI (>€2.5M), 4 as Co-I*

Conference support grants for Rocky Worlds IV in 01/26, Dutch Research Council (NWO), SRON, Netherlands Platform for Planetary Science, Leids Kerkhoven-Bosscha Fonds, PI 2025

AEThER Collaboration, Sloan Foundation, Co-I 07/25

European Research Council's (ERC) Starting Grant MagmaWorlds (€1.5M), PI 09/25

Dutch National Science Organisation (NWO) Consortium Grant (€329'065 out of €6.7M in total, *PRELIFE - Pathways, Reactions, and Environments leading to LIFE*), work package leader, Co-I 11/24

NOVA Consortium Grant (LIFE: Mid-infrared detection of biosignatures in the atmospheres of terrestrial exoplanets), PI: van der Tak (fte + cash € 249'000), Co-I 07/24

Open eScience Early Career Project PROTEUS (€134'000, *An Adaptive Numerical Framework to Simulate Lava Exoplanets*), Netherlands eScience Center, PI 12/23

Workshop grant for Rocky Worlds Discussions ($2'000), Meteoritical Society 02/23

Faculty Research Grant, Faculty of Science and Engineering (€150'000), University of Groningen, PI 12/22

Research Fellowship of the Branco Weiss Foundation (*CHF 500'000, Molten Exoplanets as a Window into the Earliest Earth*), PI 08/22

A-rated ERC Starting Grant Proposal (reserve list), European Research Council 06/22

Conference support grants for Rocky Worlds II in 01/22 (£8'000 + $5'000), Royal Astronomical Society, The Meteoritical Society, Kavli Institute for Cosmology, PI 05/21

Non-stipendiary Junior Research Fellowship, St Cross College, University of Oxford, PI 06/19

Simons Collaboration on the Origins of Life Postdoctoral Fellowship ($276'000, *Geodynamic Regimes during Planetogenesis*), Simons Foundation, PI 12/18

Early Postdoc.Mobility fellowship (*CHF 77'650, Linking Interior-Atmosphere Regimes to Planetary Accretion*), Swiss National Science Foundation, PI 12/17

Conference support grant for Water during planet formation and evolution in 02/18 (*CHF 10'000*), University of Zurich Graduate Campus, Co-I 04/17

MERAC Funding and Travel Award (*CHF 5'000*), Swiss Society for Astrophysics & Astronomy, PI 10/16

## Publications

Research-ID links: ORCID • Google Scholar • ADS • SciX • Scopus • arXiv

*84 articles in total, incl. 16 first-author, 1 single-author, 22 co-/supervised student papers*, 8 mentored postdoc papers†*
*As first author 2x Science, 1x Nature Astronomy, 3x ApJL, 1x E&PSL*

### Preprints & Submitted Journal Articles

**T Lichtenberg**, †M Attia, H Nicholls, *M Sastre, DJ Bower, *K Stuitje, *FC Pascal, L Soucasse, D Apai, P Bos, R Calder, *L Cesario, L Dang, *E Decocq, *M van Dijk, M Farhat, K Hakim, †T Kimura, *I Kisvárdai, S Krijt, Y Miguel, *I Panagiotou, *E Postolec, M Schlecker, S Seager, A Shahar, O Shorttle, PA Sossi, W van Westrenen. Super-Earth Interiors Shrink by About 10% as They Crystallise.

*Sastre M, **T Lichtenberg**, L Dang, A Piette, HS Wang, M López-Morales, T Wilson, C-É Boukaré, M Herath, N Cowan, MA Al Zaman, MG Barber, C Brinkman-Traverse, N Connors, I Crossfield, L D'Aoust, O Herbort, L Janssen, M Kervazo, O Lammert, Y Miguel, R Pierrehumbert, A Vazan, JP Wardenier, S Zieba. Thermal Evolution of Lava Planets Across System Ages: Predictions for Hell of a Survey.

*Postolec E, **T Lichtenberg**, JK Teske, H Nicholls, †M Attia, A Piette, L Dang, NL Wallack, M Plotnykov, A McGinty, S Boucher, B Peng, D Valencia. Evolutionary pathways toward survival of a thick $CO_2$- or $SO_2$-rich atmosphere on the lava world TOI-561 b.

†Attia M, **T Lichtenberg**. Atmospheric escape fractionates secondary but not primary atmospheres.

†Nicholls H, **T Lichtenberg**, B Riegler, R Calder, V Fortuin. Constraining the lives and times of exoplanets through evolutionary Bayesian retrievals.

*Panagiotou I, **T Lichtenberg**, S-M Tsai, H Nicholls. Sulfur photochemistry observationally traces mantle redox states of rocky planets.

†Nicholls H, J Krissansen-Totton, **T Lichtenberg**, L Schaefer, K Hamano, M Maurice, H Samuel, A Papesh, C Ortiz-Quintana, J Perez, Y Miguel, D Sergeev, P Baumeister, S Dash, L Janssen, J Keathley, A de Larminat, E Marcq, L Noack, H Pelissard, B Peng, *E Postolec, R Ramirez, *M Sastre, A Zorzi. Coupled atmospHere Interior modeL Intercomparison (CHILI). I. Evolutionary Modelling – Primordial Magma Oceans of Earth and Venus.

### Refereed Research Journal Articles

Zieba S, B Edwards, SN Quinn, LJ Janssen, NB Cowan, L Kreidberg, M Lopez-Morales, L Dang, L Malavolta, Y Miguel, TG Nguyen, A Vanderburg, TG Wilson, M Zilinskas, NJ Connors, IJM Crossfield, O Herbort, AL Kraus, **T Lichtenberg**, *M Sastre, JP Wardenier. The First Hubble Detection of a Secondary Eclipse from a Rocky Exoplanet: the 5.4-hour planet TOI-2431 b. *The Astrophysical Journal Letters*, in press (2026).

Turner DA, A Mortier, JG Rogers, JA Egger, AV Freckelton, **T Lichtenberg**, AAA Piette. No statistically significant evidence for a correlation between stellar and planetary composition. *Monthly Notices of the Royal Astronomical Society*, in press (2026).

Boucher S, L Dang, A McGinty, JK Teske, RT Pierrehumbert, AAA Piette, NL Wallack, A Weeks, NT Lewis, **T Lichtenberg**, M Plotnykov, *E Postolec, D Huber, TR Bedding, H Nicholls, B Peng, M Hammond, D Valencia. A Highly Reflective Atmosphere on the Lava World TOI-561b Revealed by JWST/NIRSpec Phase Curve. *The Astronomical Journal*, in press (2026).

†Nicholls H, O Shorttle, **T Lichtenberg**, *F Pascal. Beyond the mass-radius plane: Integrated radiative-convective and

interior structure simulations of the exoplanet continuum. *Monthly Notices of the Royal Astronomical Society* (2026).

*Cesario L, **T Lichtenberg**, M Min, L Noack, C Brachmann, E Alei, SP Quanz, LIFE Collaboration. Characterizing the oxidation state of rocky exoplanets with the Large Interferometer for Exoplanets (LIFE). *Monthly Notices of the Royal Astronomical Society*, in press (2026).

Williams J, S Krijt, J Drążkowska, **T Lichtenberg**. Planetesimal Formation Across the Stellar Mass Spectrum and its Influence on Exoplanet-Inherited Volatile Budgets. *Monthly Notices of the Royal Astronomical Society* (2026).

*Postolec E, **T Lichtenberg**, *H Nicholls, L Soucasse, F van der Tak. Atmospheric evolution through outgassing and escape on young molten rocky exoplanets. *Astronomy & Astrophysics*, in press (2026).

*Cesario L, **T Lichtenberg**, †M Attia, H Nicholls, *I Kisvárdai, Q Changeat. Reflation: Redox-Driven Atmospheric Inflation as Tracer of Super-Earth Geochemistry. *The Astrophysical Journal Letters* 1007, L8 (2026).

†Attia M, **T Lichtenberg**, *E Jungová, *M Sastre. PALEOS: Multiphase Equations of State and Mass-Radius Relations for Exoplanet Interiors. *Astronomy & Astrophysics*, in press (2026).

*Sastre M, **T Lichtenberg**, L Soucasse, DJ Bower, *H Nicholls, I Kamp. Geophysical and atmospheric implications of $fO_2$-dependent melting on rocky exoplanets. *Astronomy & Astrophysics* (2026).

Grenfell JL, B Taysum, I van Zelst, F Schreier, H Innes, AMS Smith, S Csizmadia, N Iro, S Rugheimer, T Kozakis, E Alei, L Noack, **T Lichtenberg**, SP Quanz, K Herbst, M Sinnhuber, A Bartenschlager, J Cabrera, H Rauer. Detectability of Atmospheric Climate and Biosignatures with the Large Interferometer for Exoplanets (LIFE) for Terrestrial-type Exoplanets. *Monthly Notices of the Royal Astronomical Society* 550, staf1878 (2026).

Calder R, O Shorttle, H Nicholls, **T Lichtenberg**, CM Guimond. Most Rocky Sub-Neptunes are Molten: Mapping the Solidification Shoreline for Gas Dwarf Exoplanets. *Monthly Notices of the Royal Astronomical Society* 549 (2026).

**T Lichtenberg**, L Schaefer, J Krissansen-Totton, Y Miguel, DE Sergeev, P Baumeister, J Cmiel, LJ Janssen, TG Nguyen, Y Miyazaki, H Nicholls, A Papesh, H Pelissard, B Peng, J Perez, E Postolec, M Sastre et al. Coupled atmospHere Interior modeL Intercomparison (CHILI) Protocol Version 1.0: A CUISINES Intercomparison Project of Magma Ocean Models. *The Planetary Science Journal* 7, 108 (2026).

*van Dijk MR, *H Nicholls, **T Lichtenberg**. Onset of habitable conditions on the Hadean Earth set by feedback between tides and greenhouse forcing. *The Planetary Science Journal* 7, 94 (2026).

Rugheimer S, E Alei, BS Konrad, B Taysum, JL Grenfell, **T Lichtenberg**, D Kitzmann, F van der Tak, SP Quanz, LIFE collaboration. The Goldilocks problem for detecting water in terrestrial planets: Constraining water abundances in the mid-IR with LIFE. *The Astrophysical Journal* 1003, 85 (2026).

*Nicholls H, **T Lichtenberg**, RD Chatterjee, CM Guimond, *E Postolec, RT Pierrehumbert. Volatile-rich evolution of molten super-Earth L 98-59 d. *Nature Astronomy* (2026).

†Kimura T, **T Lichtenberg**. Water enrichment of forming sub-Neptune envelopes limited by oxygen exhaustion. *The Astrophysical Journal* 1000, 220 (2026).

Meier TG, CM Guimond, RT Pierrehumbert, J Birkby, RD Chatterjee, CE Fisher, GJ Golabek, M Hammond, TD Komacek, **T Lichtenberg**, A McGinty, E Meier Valdés, H Nicholls, LT Parker, RJ Spaargaren, PJ Tackley. Mantle convection and nightside volcanism on lava world K2-141 b. *Monthly Notices of the Royal Astronomical Society* 547, stag390 (2026).

Nixon MC, S Sander Somers, AB Savel, J Ih, EMR Kempton, ED Young, HE Schlichting, **T Lichtenberg**, L Welbanks, W Misener, AAA Piette, NF Wogan. Magma ocean interactions can explain JWST observations of the sub-Neptune TOI-270 d. *The Astrophysical Journal* 995, 95 (2025).

Teske JK, NL Wallack, AAA Piette, L Dang, **T Lichtenberg**, M Plotnykov, RT Pierrehumbert, *E Postolec, et al. A Thick Volatile Atmosphere on the Ultra-Hot Super-Earth TOI-561 b. *The Astrophysical Journal Letters* 995, L39 (2025).

*Nicholls H, CM Guimond, HCFC Hay, RD Chatterjee, **T Lichtenberg**, RT Pierrehumbert. Self-limited tidal heating and prolonged magma oceans in the L 98-59 system. *Monthly Notices of the Royal Astronomical Society* 541, 2566-2584 (2025).

Apai D., R. Barnes, M.M. Murphy, **T. Lichtenberg**, et al. A Terminology and Quantitative Framework for Assessing the Habitability of Solar System and Extraterrestrial Worlds. *The Planetary Science Journal* 6, 165 (2025).

*Boer ID, *H Nicholls, **T. Lichtenberg**. Absence of a Runaway Greenhouse Limit on Lava Planets. *The Astrophysical Journal* 987, 172 (2025).

Schlecker M, D Apai, A Affholder, S Ranjan, R Ferrière, KK Hardegree-Ullman, **T. Lichtenberg**, S Mazevet. Bioverse: Potentially Observable Exoplanet Biosignature Patterns Under the UV Threshold Hypothesis for the Origin of Life. *The Astrophysical Journal* 987, 24 (2025).

Farhat M, P Auclair-Desrotour, G Boué, **T. Lichtenberg**, J Laskar. Tides on Lava Worlds: Application to Close-in Exoplanets and the Early Earth-Moon System. *The Astrophysical Journal* 979, 133 (2025).

Hammond M, CM Guimond, **T. Lichtenberg** et al. Reliable Detections of Atmospheres on Rocky Exoplanets with Photometric JWST Phase Curves. *The Astrophysical Journal Letters* 978, L40 (2025).

*Nicholls H, RT Pierrehumbert, **T. Lichtenberg** et al. Convective shutdown in the atmospheres of lava worlds. *Monthly Notices of the Royal Astronomical Society* 536, 2957-2971 (2025).

*Nicholls H, **T. Lichtenberg**, DJ Bower, RT Pierrehumbert. Magma ocean evolution at arbitrary redox state. *Journal of Geophysical Research: Planets* 129, 2024JE008576 (2024).

*Cesario L, **T. Lichtenberg** et al. Large Interferometer For Exoplanets (LIFE). XIV. Finding terrestrial protoplanets in

the galactic neighborhood. *Astronomy & Astrophysics* 692, A172 (2024).

Eatson J. W., RJ Parker, **T. Lichtenberg**. Towards a unified injection model of short-lived radioisotopes in N-body simulations of star-forming regions. *The Astrophysical Journal* 977, 13 (2024).

†Meier T. G., D.J. Bower, **T. Lichtenberg**, M. Hammond, P.J. Tackley, R.T. Pierrehumbert, et al. Geodynamics of super-Earth GJ 486b. *Journal of Geophysical Research: Planets* 129, e2024JE008491 (2024).

†Eatson J. W., **T. Lichtenberg**, R. J. Parker, T. V. Gerya. Devolatilization of extrasolar planetesimals by $^{60}$Fe and $^{26}$Al heating. *Monthly Notices of the Royal Astronomical Society* 528, 6619-6630 (2024).

Shorttle O., S. Jordan, H. Nicholls, **T. Lichtenberg**, D. J. Bower. Distinguishing oceans of water from magma on mini-Neptune K2-18b. *The Astrophysical Journal Letters* 962, L8 (2024).

Schlecker S., D. Apai, **T. Lichtenberg**, G. Bergsten, A. Salvador, K. K. Hardegree-Ullman. Bioverse: The Habitable Zone Inner Edge Discontinuity as an Imprint of Runaway Greenhouse Climates on Exoplanet Demographics. *The Planetary Science Journal* 5, 3 (2024).

Patel M., C. K. M. Polius, M. Ridsdill-Smith, **T. Lichtenberg**, R. Parker. Photoevaporation versus enrichment in the cradle of the Sun. *Monthly Notices of the Royal Astronomical Society* 525, 2399-2410 (2023).

*Meier T. G., D. J. Bower, **T. Lichtenberg**, M. Hammond, P. J. Tackley. Interior dynamics of super-Earth 55 Cancri e. *Astronomy & Astrophysics* (2023).

Piette A. A. A., P. Gao, K. Brugman, A. Shahar, **T. Lichtenberg**, F. Miozzi, P. Driscoll. Rocky world? Observability of low-density lava world atmospheres. *The Astrophysical Journal* 954, 29 (2023).

Parker R. J., **T. Lichtenberg**, M. Patel, C. K. M. Polius, M. Ridsdill-Smith. Short-lived radioisotope enrichment in star-forming regions from stellar winds and supernovae. *Monthly Notices of the Royal Astronomical Society* 521, 4838-4851 (2023).

Janson M., J. Patel, S. C. Ringqvist, C. Lu, I. Rebollido, **T. Lichtenberg**, A. Brandeker, D. Angerhausen, L. Noack. Imaging of exocomets with infrared interferometry. *Astronomy & Astrophysics* 671, A114 (2023).

Stammler S. M., **T. Lichtenberg**, J. Drążkowska, T. Birnstiel. Leaky dust traps: How fragmentation impacts dust filtering by planets. *Astronomy & Astrophysics Letters* 670, L5 (2023).

Bonsor A., **T. Lichtenberg**, J. Drążkowska, A. M. Buchan. Rapid formation of exoplanetesimals revealed by white dwarfs. *Nature Astronomy* 7, 39-48 (2023).

*Graham R. J., **T. Lichtenberg**, R. T. Pierrehumbert. $CO_2$ ocean bistability on terrestrial exoplanets. *Journal of Geophysical Research: Planets* 127, 2022JE007456 (2022).

**Lichtenberg T.**, M. S. Clement. Reduced late bombardment on rocky exoplanets around M-dwarfs. *The Astrophysical Journal Letters* 938, L3 (2022).

Curry A., A. Bonsor, **T. Lichtenberg**, O. Shorttle. Prevalence of short-lived radioactive isotopes across exoplanetary systems inferred from polluted white dwarfs. *Monthly Notices of the Royal Astronomical Society* 515, 395-406 (2022).

Dannert F. et al. (incl. **T. Lichtenberg**) Large Interferometer For Exoplanets (LIFE): II. Signal simulation, signal extraction and fundamental exoplanet parameters from single epoch observations. *Astronomy & Astrophysics* 664, A22 (2022).

Quanz S. P. et al. (incl. **T. Lichtenberg**) Large Interferometer For Exoplanets (LIFE): I. Improved exoplanet detection yield estimates for a large mid-infrared space-interferometer mission. *Astronomy & Astrophysics* 664, A21 (2022).

Jin Z., M. Bose, **T. Lichtenberg**, G. D. Mulders. New evidence for wet accretion of inner solar system planetesimals from meteorites Chelyabinsk and Benenitra. *The Planetary Science Journal*, 2, 244 (2021).

Tsai S.-M., H. Innes, **T. Lichtenberg**, J. Taylor, M. Malik, K. Chubb, R. T. Pierrehumbert. Inferring Shallow Surfaces on Sub-Neptune Exoplanets with JWST. *The Astrophysical Journal Letters* 922, L27 (2021).

Dorn C., **T. Lichtenberg**. Hidden water in magma ocean exoplanets. *The Astrophysical Journal Letters* 922, L4 (2021). (summary video)

*Graham R. J., **T. Lichtenberg**, R. Boukrouche*, R. T. Pierrehumbert. A multispecies pseudoadiabat for simulating condensable-rich exoplanet atmospheres. *The Planetary Science Journal* 2, 207 (2021). (summary video)

*Boukrouche R., **T. Lichtenberg**, R. T. Pierrehumbert. Beyond runaway: initiation of the post-runaway greenhouse state on rocky exoplanets. *The Astrophysical Journal* 919, 130 (2021). (summary video)

**Lichtenberg T.** Redox hysteresis of super-Earth exoplanets from magma ocean circulation. *The Astrophysical Journal Letters* 914, L4 (2021). (summary blog, summary video)

**Lichtenberg T.**, S. Krijt. System-level fractionation of carbon from disk and planetesimal processing. *The Astrophysical Journal Letters* 913, L20 (2021). (summary blog, summary video)

*Meier T. G., D. J. Bower, **T. Lichtenberg**, P. J. Tackley, B.-O. Demory. Hemispheric tectonics on super-Earth LHS 3844b. *The Astrophysical Journal Letters* 908, L48 (2021). (summary video)

**Lichtenberg T.**, D. J. Bower, M. Hammond, R. Boukrouche*, P. Sanan, S.-M. Tsai, R. T. Pierrehumbert. Vertically resolved magma ocean-protoatmosphere evolution: $H_2$, $H_2O$, $CO_2$, $CH_4$, CO, $O_2$, and $N_2$ as primary absorbers. *Journal of Geophysical Research: Planets* 126, e2020JE006711 (2021). (summary blog, summary video)

**Lichtenberg T.**, J. Drążkowska, M. Schönbächler, G. J. Golabek, T. O. Hands. Bifurcation of planetary building blocks during Solar System formation. *Science* 371, 365–370 (2021). (summary blog, summary video)

**Lichtenberg T.**, G. J. Golabek, R. Burn, M. R. Meyer, Y. Alibert, T. V. Gerya, C. Mordasini. A water budget dichotomy of rocky protoplanets from $^{26}$Al heating. *Nature Astronomy* 3, 307–313 (2019). (summary blog)

**Lichtenberg T.**, T. Keller, R. F. Katz, G. J. Golabek, T. V. Gerya. Magma ascent in planetesimals: control by grain size. *Earth and Planetary Science Letters* 507, 154–165 (2019). (blog post)

*Bonati I., **T. Lichtenberg**, D. J. Bower, M. Timpe, S. P. Quanz. Direct imaging of molten protoplanets in nearby young stellar associations. *Astronomy & Astrophysics* 621, A125 (2019). (blog post)

**Lichtenberg T.**, G. J. Golabek, C. P. Dullemond, M. Schönbächler, T. V. Gerya, M. R. Meyer. Impact splash chondrule formation during planetesimal recycling. *Icarus* 302, 27–43 (2018). (blog post)

Hunt A. C., D. L. Cook, **T. Lichtenberg**, P. M. Reger, M. Ek, G. J. Golabek, M. Schönbächler. Late metal-silicate separation on the IAB parent asteroid: Constraints from combined W and Pt isotopes and thermal modelling. *Earth and Planetary Science Letters* 482, 490–500 (2018).

Parker R. J., **T. Lichtenberg**, S. P. Quanz. Was Planet 9 captured in the Sun's natal star-forming region? *Monthly Notices of the Royal Astronomical Society: Letters* 472, L75–L79 (2017).

**Lichtenberg T.**, R. J. Parker, M. R. Meyer. Isotopic enrichment of forming planetary systems from supernova pollution. *Monthly Notices of the Royal Astronomical Society* 462, 3979–3992 (2016). (blog post)

**Lichtenberg T.**, G. J. Golabek, T. V. Gerya, M. R. Meyer. The effects of short-lived radionuclides and porosity on the early thermo-mechanical evolution of planetesimals. *Icarus* 274, 350–365 (2016). (blog post)

**Lichtenberg T.**, D. R. G. Schleicher. Modeling gravitational instabilities in self-gravitating protoplanetary disks with adaptive mesh refinement techniques. *Astronomy & Astrophysics* 579, A32 (2015).

### Refereed Review Articles

**Lichtenberg T**, O Shorttle, J Teske, EMR Kempton. Constraining exoplanet interiors using observations of their atmospheres. *Science* 390, eads3360 (2025).

**Lichtenberg T.**, Y. Miguel. Super-Earths and Earth-like Exoplanets. *Treatise on Geochemistry*, 3rd edition, Vol. 7, 51-112 (2025).

Suer T.-A., C. Jackson, D. S. Grewal, C. Dalou, **T. Lichtenberg**. The distribution of volatile elements during rocky planet formation. *Frontiers in Earth Science* 11:1159412 (2023).

**Lichtenberg T.**, L. K. Schaefer, M. Nakajima, R. Fischer. Geophysical Evolution During Rocky Planet Formation. *Protostars and Planets VII*, ASP Conference Series, Vol. 534, 907 (2023).

### Perspectives and Collaboration Papers

Hu R. et al. (incl. **T. Lichtenberg**). Identifying rocky planets and water worlds among sub-Neptune-sized exoplanets with the Habitable Worlds Observatory. arXiv preprint (2025).

Rauer H. et al. (incl. **T. Lichtenberg**). The PLATO mission. *Experimental Astronomy* 59, 26 (2025).

Menti F. et al. (incl. **T. Lichtenberg**). Database of Candidate Targets for the LIFE Mission. *Research Notes of the AAS*, 8, 267 (2024).

TRAPPIST-1 JWST Community Initiative (incl. **T. Lichtenberg**) A roadmap for the atmospheric characterization of terrestrial exoplanets with JWST. *Nature Astronomy Perspective*, Vol. 8, 810-818 (2024).

Helled R. et al. (incl. **T. Lichtenberg**). Ariel Planetary Interiors White Paper. *Experimental Astronomy* 53, 323-356 (2021).

Tinetti G. et al. (incl. **T. Lichtenberg**). Ariel: Enabling planetary science across light-years. *ESA Ariel Definition Study Report* (2021).

### Instrumentation & Methods

*Nicholls H., R.T. Pierrehumbert, **T. Lichtenberg**. AGNI: A radiative-convective model for lava planet atmospheres. *Journal of Open Source Software* 10, 7726 (2025).

Glauser A. M., S. P. Quanz, J. Hansen et al. (incl. **T. Lichtenberg**). The Large Interferometer For Exoplanets (LIFE): a space mission for mid-infrared nulling interferometry. *Proceedings of the SPIE* 13095, 130951D (2024).

### White Papers (not in publication count)

Meadows V. et al. (incl. **T. Lichtenberg**) Community Report from the Biosignatures Standards of Evidence Workshop. Workshop Report (2022).

Khullar G. et al. (incl. **T. Lichtenberg**). Astrobites as a Community-led Model for Education, Science Communication, and Accessibility in Astrophysics. Astro2020: NASA Decadal Survey on Astronomy and Astrophysics, *BAAS*, 51, 7, 230 (2019).

Quanz S. P. et al. (incl. **T. Lichtenberg**). Atmospheric characterization of terrestrial exoplanets in the mid-infrared: biosignatures, habitability & diversity. *ESA Voyage 2050* White Paper (2019).

Line M. et al. (incl. **T. Lichtenberg**). The Importance of Thermal Emission Spectroscopy for Understanding Terrestrial Exoplanets. Astro2020: NASA Decadal Survey, *BAAS*, 51, 3, 271 (2019).

Lyra W. et al. (incl. **T. Lichtenberg**). Planet formation — The case for large efforts on the computational side. Astro2020: NASA Decadal Survey, *BAAS*, 51, 3, 271 (2019).

## Conference and Institute Seminar Presentations

*168 presentations in total, incl. 81 invited*‡

### Conferences and Workshops

| | | |
|---|---|---|
| BUFFET-6 (virtual) ‡ | Talk | 09/26 |
| Europlanet Science Congress, The Hague, NL | Talk | 09/26 |
| Layers of Understanding: Model Intercomparisons of Exoplanet Interiors, Heidelberg, DE ‡ | Talk | 04/26 |
| International Conference on Exoplanets and Planet Formation, Shanghai, CN ‡ | Talk | 12/25 |
| Planetary Formation and Exoplanets in the ELT Era, Garching, DE | Talk | 11/25 |
| CHILI: Coupled atmospHere Interior modeL Intercomparison, Leiden, NL | Talk | 09/25 |
| Frontiers in Nanoscience CeNS/CRC 392, Venice, IT ‡ | Talk | 09/25 |
| BUFFET-5, Bordeaux, FR ‡ | Talk | 09/25 |
| Europlanet Science Congress, Helsinki, FI ‡ | Talk | 09/25 |
| ESA Ariel Consortium Meeting, Leiden, NL | Talk | 04/24 |
| Future Role of FIR-Submm Space Observations, Leiden, NL ‡ | Talk | 04/24 |
| NOVA @ 25, The Hague, NL ‡ | Talk | 12/24 |
| Origins Federation Conference, Cambridge, UK ‡ | Talk | 09/24 |
| National Astronomers' Conference, NL ‡ | Talk | 05/24 |
| The Geoscience of Exoplanets: Going Beyond Habitability, International Space Science Institute, Bern, CH ‡ | Talk | 04/24 |
| Challenges in the field of exoplanets and the search for life, Leiden, NL ‡ | Talk | 02/24 |
| Ringberg Conference: Density Matters, Ringberg Castle, Kreuth, DE ‡ | Talk | 02/24 |
| Atmospheric and Interior connection in rocky EXOplanets and what we can learn from the Solar System, JPL Caltech, Pasadena, US (virtual) ‡ | Talk | 08/23 |
| Exoclimes VI, Exeter, UK | Talk | 06/23 |
| Molecular Origins of Life, Munich, DE (virtual) ‡ | Talk | 06/23 |
| Gordon Research Conference – Origins of Solar Systems, South Hadley, US ‡ | Talk | 06/23 |
| European Geosciences Union (EGU) General Assembly, Vienna, AUT ‡ | Talk | 04/23 |
| Protostars and Planets VII, Kyoto, JP ‡ | Talk | 04/23 |
| ESA ESLAB – Untrstanding Planets in the Solar System and Beyond, Noordwijk, NL | Talk | 03/23 |
| Fundamentals of the Universe Symposium, Groningen, NL | Talk | 02/23 |
| Life in the Universe, Physics of Living Systems, Sofia, BG ‡ | Talk | 10/22 |
| The Sun and Planetary Systems, Center for Space Science, NYUAD, UAE ‡ | Talk | 10/22 |
| 10th Workshop on High Pressure, Planetary and Plasma Physics, Royal Observatory of Belgium, Brussels, BE | Talk | 09/22 |
| 3rd Workshop on Giant Collisions and their Effects on the Thermochemical Evolution of Planets, DLR Berlin, DE | Talk | 09/22 |
| (Exo)Planet Diversity, Formation and Evolution, FU Berlin, DE | Talk | 09/22 |
| Latsis Symposium on The Origin and Prevalence of Life, ETH Zurich, CH | Talk | 09/22 |
| International Astronomical Union General Assembly, FM10: Synergy of Small Telescopes, and Large Surveys for Solar System and Exoplanetary Bodies Research, Busan, KR | Talk | 08/22 |
| International Astronomical Union General Assembly, Division F: Planetary Systems and Astrobiology, Busan, KR ‡ | Talk | 08/22 |
| Annual Meeting of the European Astronomical Society, Valencia, ESP ‡ | Talk | 06/22 |
| AGU21: Multidisciplinary Perspectives on Terrestrial Worlds (virtual) | Talk | 12/21 |
| AGU21: A Journey Into Planetary Interiors (virtual) | Talk | 12/21 |
| ESA PLATO Atmospheres workshop 2021 (virtual) | Talk | 12/21 |

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| The Volatile Content of Planets that Form Early, Lorentz Center workshop, Leiden, NE (virtual) ‡ | Talk | 12/21 |
| Ariel Interiors working group meeting 2021, Zurich, CH (virtual) | Talk | 11/21 |
| PLATO Mission Conference 2021, Berlin, DE (virtual) | Talk | 10/21 |
| Ringberg 2021: Formation of the Solar System, Kreuth, DE ‡ | Talk | 10/21 |
| EPSC 2021, EXO3: Exoplanet observations, modelling and experiments, virtual, EU | Talk | 09/21 |
| EPSC 2021, TP3: Late accretion processes, virtual, EU ‡ | Talk | 09/21 |
| EPSC 2021, EXO1: Zooming In On Planet Formation, EU (virtual) | Talk | 09/21 |
| Astrobiology Graduate Conference 2021, Tokyo, JP (virtual) | Talk | 09/21 |
| German Physical Society Meeting, session Exoplanets & Astrobiology, virtual, DE | Talk | 09/21 |
| Molecular Origins of Life 2021, Munich, DE (virtual) | Poster | 08/21 |
| Simons Collaboration on the Origins of Life Investigators Meeting, New York, US (virtual) | Talk | 07/21 |
| Goldschmidt 21, Chemical geodynamics throughout the Solar System, Lyon, FR (virtual) ‡ | Talk | 07/21 |
| STScI SS21, Towards the Comprehensive Characterization of Exoplanets, STScI, US (virtual) | Talk | 04/21 |
| STScI Spring Symposium 2021, Science Writers' Workshop, STScI, US (virtual) ‡ | Talk | 04/21 |
| Habitable Worlds 21, NASA NExSS, NExScI, US (virtual) ‡ | Talk | 02/21 |
| Origins 2021, Origins Center, NL (virtual) | Talk | 01/21 |
| AGU: Comparative Planetology Throughout the Solar System, San Francisco, US (virtual) | Poster | 12/20 |
| Five years after HL Tau: a new era in planet formation, ESO, CL (virtual) | Talk | 12/20 |
| ESO Threats from the Surroundings, Garching, DE (virtual) | Poster | 11/20 |
| Exoplanet Demographics 2020, Pasadena, US (virtual) | Poster | 11/20 |
| Simons Collaboration on the Origins of Life Annual Symposium, New York, US (virtual) | Talk | 11/20 |
| EPSC 2020, EXO3: From Disks to Planets and their Atmospheres, virtual, EU | Talk | 09/20 |
| EPSC 2020, TP3: Coupled planet formation and evolution, EU (virtual) | Talk | 09/20 |
| Exoplanets III, Heidelberg, DE (virtual) | Poster | 07/20 |
| Molecular Origins of Life 2020, Munich, DE (virtual) | Poster | 07/20 |
| MIAPP virtual workshop: Final Orbit Assembly, Garching, DE (virtual) | Talk | 06/20 |
| What Makes a Planet Uninhabitable?, Chicago, US (virtual) ‡ | Talk | 05/20 |
| Gordon Research Seminar & Conference – Origins of Life, Galveston, US | Talk | 01/20 |
| Rocky Worlds: from the Solar System to Exoplanets, Cambridge, UK | Talk | 01/20 |
| Exoplanet Vision 2050, Konkoly Observatory, Hun Academy of Sciences, Budapest, HUN ‡ | Talk | 11/19 |
| RU Transition Disks meeting, Center for Advanced Studies, LMU Munich, Munich, DE | Talk | 11/19 |
| EPSC-DPS 2019, TP16: Collisions from small bodies to planetary scales, Geneva, CH | Talk | 09/19 |
| Annual Meeting of the German Astronomical Society 2019, Stuttgart, DE ‡ | Talk | 09/19 |
| EPSC-DPS 2019, TP14: Coupled planet formation and evolution, Geneva, CH | Talk | 09/19 |
| EPSC-DPS 2019, EXO2: Exoplanet observations, modelling and experiments, Geneva, CH | Talk | 09/19 |
| Extreme Solar Systems IV, Reykjavík, IS | Poster | 08/19 |
| Exoclimes V, Oxford, UK | Poster | 08/19 |
| Gordon Research Conference – Origins of Solar Systems, South Hadley, USA | Poster | 06/19 |
| From Stars to Planets II, Gothenburg, SE | Talk | 06/19 |
| UK Exoplanet Community Meeting 2019, London, UK | Talk | 04/19 |
| Planet Formation and Evolution 2019, Rostock, DE | Talk | 02/19 |
| 2nd Oxford Meeting on Planets, Oxford, UK | Talk | 12/18 |
| European Astrobiology Network Association 2018, Berlin, DE ‡ | Talk | 09/18 |
| European Planetary Science Congress 2018, session EXO1, Berlin, DE | Poster | 09/18 |
| EPSC 2018, EXO2: Formation and Dynamical Evolution of Planetary Systems, Berlin, DE | Talk | 09/18 |
| EPSC 2018, SB4: Asteroids and parent bodies of meteorites, Berlin, DE | Talk | 09/18 |
| Exoplanets II, Cambridge, UK | Talk | 07/18 |

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| UK Exoplanet Community Meeting, Oxford, UK | Talk | 03/18 |
| Water during planet formation and evolution, Zurich, CH | Poster | 02/18 |
| NCCR PlanetS General Assembly 2018, Grindelwald, CH | Poster | 01/18 |
| The Origins of Volatiles in Habitable Planets, Ann Arbor, US | Poster | 10/17 |
| Planet Formation and Evolution 2017, Jena, DE | Poster | 09/17 |
| Gordon Research Conference 'Origins of Solar Systems', South Hadley, USA | Poster | 06/17 |
| Gordon Research Seminar 'Origins of Solar Systems', South Hadley, USA | Talk | 06/17 |
| Accretion, Differentiation and Early Evolution of the Terrestrial Planets, Nice, FR | Talk | 05/17 |
| Chondrules and the Protoplanetary Disk, London, UK | Talk | 02/17 |
| Formation of the Solar System and the Origin of Life, Leiden, NL | Talk | 02/17 |
| German-Swiss Geodynamics Workshop, Lichtenfels, DE | Poster | 09/16 |
| New Directions in Planet Formation, Leiden, NL | Talk | 07/16 |
| Origin and Evolution of Plate Tectonics, Ascona, CH | Poster | 07/16 |
| Exoplanets I, Davos, CH | Poster | 07/16 |
| 4$^{th}$ Sant Cugat Forum on Astrophysics, Sant Cugat, ESP | Talk | 04/16 |
| Planet Formation and Evolution 2016, Duisburg-Essen, DE | Talk | 03/16 |
| The Delivery of Water to Proto-planets, Planets and Satellites, ISSI Bern, CH | Talk | 01/16 |
| National Centre for Competence in Research PlanetS General Assembly, Grindelwald, CH | Talk | 01/16 |
| Gordon Research Conference 'Origins of Solar Systems', South Hadley, USA | Poster | 07/15 |
| NCCR PlanetS General Assembly 2015, Anzère, CH | Poster | 01/15 |
| Planet Formation and Evolution, Kiel, DE | Poster | 09/14 |
| Annual workshop of the German NSF Graduiertenkolleg 1351, Salzgitter, DE | Talk | 09/13 |

### Colloquia and Seminar Talks

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| Institute of Astronomy, KU Leuven, Leuven, BE ‡ | Talk | 10/26 |
| Globe Institute, Centre for Star and Planet Formation, Copenhagen, DK ‡ | Talk | 02/26 |
| Department of Astrophysics, Radboud University, Nijmegen, NL ‡ | Talk | 03/25 |
| Anton Pannekoek Institute for Astronomy, University of Amsterdam, Amsterdam, NL ‡ | Talk | 11/24 |
| Max Planck Institute for Solar System Research, Göttingen, DE ‡ | Talk | 05/24 |
| Graz-Vienna Exoplanet Scientist Meeting, Graz, AUT ‡ | Talk | 05/24 |
| Austrian Academy of Sciences Space Research Institute, Graz, AUT ‡ | Talk | 05/24 |
| Alien Earths Collaboration Annual All-Hands Meeting, Tucson, USA ‡ | Talk | 04/24 |
| University of Groningen, Department of Physics, Groningen, NL ‡ | Talk | 02/24 |
| SRON Netherlands Institute for Space Research, Leiden, NL ‡ | Talk | 10/23 |
| Royal Astronomical Society Ordinary Meeting, London, UK ‡ | Talk | 10/23 |
| Leiden University Observatory, Leiden, NL ‡ | Talk | 09/23 |
| Harvard University, Department of Earth and Planetary Sciences, Cambridge, US ‡ | Talk | 06/23 |
| ASTRON – Netherlands Institute for Radio Astronomy, Dwingeloo, NL ‡ | Talk | 02/23 |
| Shanghai Astronomical Observatory, Chinese Academy of Sciences, Shanghai, CN ‡ | Talk | 11/22 |
| California Institute of Technology, Division of Geological and Planetary Sciences, Yung Lunch Seminar, Pasadena, USA ‡ | Talk | 07/22 |
| University of Vienna, Department of Astrophysics, Vienna, AUT ‡ | Talk | 06/22 |
| AEThER Collaboration Seminar, Carnegie Earth and Planets Laboratory, Washington DC, USA ‡ | Talk | 06/22 |
| Munich Institute for Astro-, Particle- and BioPhysics, Planet Formation: From Dust Coagulation to Final Orbit Assembly, Garching, DE ‡ | Talk | 06/22 |
| University of Groningen, Kapteyn Astronomical Institute, Groningen, NL ‡ | Talk | 05/22 |
| Free University of Berlin, Department of Earth Sciences, Berlin, DE ‡ | Talk | 04/22 |
| Imperial College, Department of Earth Sciences, London, UK ‡ | Talk | 03/22 |

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| European Astrobiology Institute Seminar, Strasbourg, FR, (virtual) ‡ | Talk | 03/22 |
| Washington University St. Louis, Physics Colloquium, St. Louis, US (virtual) ‡ | Talk | 03/22 |
| University of Rochester, Earth and Environmental Sciences, Rochester, US (virtual) ‡ | Talk | 02/22 |
| University of Arizona, Origins Seminar, Tucson, US (virtual) ‡ | Talk | 02/22 |
| Purdue University, Earth, Atmospheric and Planetary Sciences, West Lafayette, US (virtual) ‡ | Talk | 02/22 |
| Heidelberg Initiative for the Origins of Life Colloquium, Max Planck Institute for Astronomy, Heidelberg, DE ‡ | Talk | 11/21 |
| St Andrews Centre for Exoplanet Science, St Andrews, UK (virtual) ‡ | Talk | 07/21 |
| Dominion Radio Astrophysical Observatory (DRAO), Penticton, CA (virtual) | Talk | 06/21 |
| TRR-170/Planetology colloquium, Münster, DE (virtual) ‡ | Talk | 06/21 |
| NExSS Quantitative Habitability science working group, Tucson, US (virtual) ‡ | Talk | 05/21 |
| Université de Lorraine, Petrographic and Geochemical Research Center, Nancy, FR (virtual) ‡ | Talk | 03/21 |
| University of Zurich, Theoretical Astrophysics & Computational Science, Zurich, CH (virtual) ‡ | Talk | 03/21 |
| Lunar and Planetary Institute, USRA, Houston, US (virtual) | Talk | 03/21 |
| Max Planck Institute for Astronomy, Atmospheric Physics of Exoplanets, Heidelberg, DE (virtual) ‡ | Talk | 03/21 |
| Carnegie Institution of Science, Earth & Planets Laboratory, Washington DC, US (virtual) ‡ | Talk | 02/21 |
| University of Bern, Theoretical Astrophysics and Planetary Science, Bern, CH (virtual) ‡ | Talk | 02/21 |
| University of Bristol, Isotope Geochemistry, Bristol, UK (virtual) ‡ | Talk | 02/21 |
| University of Cambridge, Planetary Geochemistry, Cambridge, UK (virtual) ‡ | Talk | 02/21 |
| São Paulo State University, Orbital Dynamics and Planetology, Sao Paulo, BR (virtual) ‡ | Talk | 10/20 |
| NIT Rourkela, Department of Physics and Astronomy, Rourkela, IND (virtual) ‡ | Talk | 10/20 |
| Berkeley CIPS / UCLA joint planetary science seminar, Berkeley, US (virtual) ‡ | Talk | 09/20 |
| Universidad de Concepción, Departamento de Astronomía, Concepción, CL (virtual) ‡ | Talk | 06/20 |
| University of Arizona, Steward Observatory, Tucson, US | Talk | 01/20 |
| ETH Zurich, Department of Earth Sciences, Zurich, CH ‡ | Talk | 11/19 |
| Imperial College, Blackett Laboratory, London, UK | Talk | 11/19 |
| McMaster University, Origins Institute, Hamilton, CA | Talk | 11/19 |
| ETH Zurich, Institute for Particle Physics and Astrophysics, Zurich, CH | Talk | 05/19 |
| German Aerospace Center (DLR) Berlin, Berlin, DE ‡ | Talk | 04/19 |
| University of Oxford, Department of Earth Sciences, Oxford, UK | Talk | 02/19 |
| University of Cambridge, Institute of Astronomy, Cambridge, UK ‡ | Talk | 05/18 |
| ETH Zurich, Institute of Geochemistry and Petrology, Zurich, CH ‡ | Talk | 04/18 |
| Stanford University, Department of Earth Sciences, Stanford, US | Talk | 10/17 |
| University of Sheffield, Department of Physics & Astronomy, Sheffield, UK | Talk | 08/17 |
| University of Oxford, Department of Earth Sciences, Oxford, UK | Talk | 07/17 |
| University of Bern, Center for Space and Habitability, CH ‡ | Talk | 06/17 |
| Hungarian Academy of Sciences, Konkoly Observatory, Budapest, HUN ‡ | Talk | 03/17 |
| University of Copenhagen, Centre for Star and Planet Formation, Copenhagen, DK ‡ | Talk | 02/17 |
| Liverpool John Moores University, Astrophysics Research Institute, Liverpool, UK | Talk | 07/16 |
| University of Oxford, Department of Earth Sciences, Oxford, UK | Talk | 06/16 |
| ETH Zurich, Institute of Geophysics, Zurich, CH | Talk | 12/15 |

## Student Mentoring

*39 supervised students in total, incl. 7 PhD, 9 MSc, 16 BSc theses, 7 projects/internships*

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| Visiting PhD research project “Impact re-triggering of protoplanet magma oceans”, Anna Grace Ulses (University of Washington, Seattle), Kapteyn Institute | 05/26– |

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| BSc thesis “Radiation-recombination-limited hydrodynamic escape: conditions for limiting atmospheric escape on sub-Neptunes and super-Earths” (co-supervised with Mara Attia), Malina Ovesen, Kapteyn Institute | 04/26–07/26 |
| BSc thesis “The influence of atmospheric composition and photodissociation on Jeans escape from exoplanets” (co-supervised with Mara Attia), Ioana Balint, Kapteyn Institute | 04/26–07/26 |
| BSc thesis “Impact-driven escape on super-Earth exoplanets” (co-supervised with Mara Attia), Renske Beuker, Kapteyn Institute | 04/26– |
| BSc thesis “Diffusion-limited atmospheric escape of hydrogen from exoplanets” (co-supervised with Mara Attia), Viesturs Streļčs, Kapteyn Institute | 04/26–07/26 |
| PhD thesis “Prebiotic climate and ocean levels of the Hadean Earth”, Emeline Decocq, Kapteyn Institute | 09/25– |
| PhD thesis “Interior-atmosphere evolution on low-mass exoplanets”, Imre Kisvárdai, Kapteyn Institute | 09/25– |
| MSc thesis “Developing an optimal observational strategy for ultra-short period super-Earths with the Ariel survey”, Lorenzo Cesario, Kapteyn Institute | 09/25–07/26 |
| MSc thesis “Ocean formation on sub-Neptune exoplanets”, Ema Jungová, Kapteyn Institute | 09/25– |
| MSc thesis “Sulfur Photochemistry in the Coupling between Interior Properties and Atmospheric Compositions of Hot Rocky Exoplanets”, Ioannis Panagiotou, Kapteyn Institute | 09/25–07/26 |
| BSc thesis “Tidally supported Hadean Magma Oceans under Secondary Atmospheres: Implications for the Production of Life’s Building Blocks” & further research projects, Marijn van Dijk, Kapteyn Institute | 04/25–07/25 |
| BSc thesis “Timescale of ocean formation and their composition on Earth-like exoplanets”, Anvel de Sainte Marie d’Agneaux, Kapteyn Institute | 04/25–07/25 |
| BSc thesis “Modelling the Interior Structure of Rocky Exoplanets with Zalmoxis”, Flavia Pascal, Kapteyn Institute | 04/25–07/25 |
| BSc thesis “The Cosmic Shoreline: Modelling Bare-rock Surfaces and Atmospheres on Rocky M-dwarf Exoplanets” & further research projects, Karen Stuitje, RUG Physics department | 04/25–07/25 |
| Internship & MSc thesis “UV flux of prebiotic atmospheres at varying redox conditions”, Bruna Fena (FU Berlin MSc student), Kapteyn Institute | 04/25– |
| MSc thesis “Probing abiotic planetary oxygenation with LIFE”, Sanne Borgmann, Kapteyn Institute | 01/25– |
| BSc Honours College research project “Exoplanet interior structure modelling”, Flavia Pascal, Kapteyn Institute | 01/25–04/25 |
| MSc thesis “Forging Planets: Exploring the Magma Ocean Geochemistry of super-Earths and sub-Neptunes”, Jorick Lania, Kapteyn Institute | 10/24–07/25 |
| BSc thesis “The under-dense region between super-Earths and sub-Neptunes”, Jessica Helmerhorst, Kapteyn Institute | 04/24–07/24 |
| BSc thesis “Rock-vapour species in lava exoplanet atmospheres”, Björn Koops, Kapteyn Institute | 04/24–07/24 |
| BSc thesis “Redox-dependency of the runaway greenhouse limit”, Iris Boer, Kapteyn Institute | 04/24–07/24 |
| BSc semester project “Characterising exoplanet surface composition with LIFE”, Flavia Pascal, Kapteyn Institute | 11/23–06/24 |
| MSc thesis “Magma oceanography of GJ 367b”, Lotte Bartels, Kapteyn Institute | 10/23–06/24 |
| PhD thesis “Evolution of magma ocean exoplanet atmospheres”, Emma Postolec, Kapteyn Institute | 10/23– |
| PhD thesis “Evolution of magma ocean exoplanet interiors”, Mariana V. Sastre, Kapteyn Institute | 09/23– |
| PhD thesis “What happened to rocky planets?”, Harrison Nicholls, Oxford Atmospheric Physics, co-supervisor | 04/23–01/26 |
| BSc thesis “Distinguishing surface features on rocky exoplanets”, Lucas Priolet, Kapteyn Institute | 04/23–07/23 |
| BSc thesis “The prevalence of steam atmospheres on hot rocky exoplanets”, Boyd van der Plaat, Kapteyn Institute | 04/23–07/23 |
| BSc thesis “Detecting Cooling Protoplanets with Next-Generation Surveys”, Lorenzo Cesario, Kapteyn Institute | 04/23–07/23 |
| BSc thesis “Volcanic climate change on hot terrestrial exoplanets”, Marijn Smorenburg, Kapteyn Institute | 04/23–07/23 |
| MSc thesis “Hydrology of Planetesimals”, Beat Hubmann, Institute of Geophysics, ETH Zurich, co-supervision | 02/22–10/22 |
| Semester project “Investigating the surface composition of airless, terrestrial exoplanets with the Large Interferometer for Exoplanets”, Gustav Pap, ETH Zurich, co-supervision | 02/21–06/21 |

PhD research projects "A multispecies pseudoadiabat for simulating condensable-rich exoplanet atmospheres", Robert J. Graham, Oxford Atmospheric Physics, co-supervision — 08/20–09/22

PhD research project "Beyond runaway: initiation of the post-runaway greenhouse state on rocky exoplanets", Ryan Boukrouche, Oxford Atmospheric Physics, co-supervision — 01/20–09/22

Pastoral support and mentoring for 16 Oxford M.Sc. and D.Phil. (Ph.D.) students from the natural sciences (Engineering, Computer Science, Medicine, Physics, Chemistry), St Cross College, University of Oxford — 10/19–09/22

Semester project "Influence of silicate rheology on iron core formation", Madeleine Müller, Institute of Geophysics, ETH Zurich, co-supervision — 03/20–06/20

BSc thesis "Percolative core formation in terrestrial planetesimals and protoplanets", Madeleine Müller, Institute of Geophysics, ETH Zurich, co-supervision — 12/18–06/19

MSc thesis "Quantitative Predictions for the Observability of Protoplanetary Collisions", Irene Bonati, Institute of Geophysics, ETH Zurich, co-supervision — 12/16–09/17

Semester project "2D Numerical Modeling of Pebble Accretion Influence on Planetesimal Evolution", Fabio Luchsinger, Institute of Geophysics, ETH Zurich, co-supervision — 08/16–02/17

## Lectures and Courses

*16 courses in total, incl. 10 as lecturer, 6 as teaching/lab assistant*

*Exoplanets*, MSc course, Kapteyn Institute, lecturer (∼20 students) — 02/26–04/26

*Planetary Systems*, BSc course, Kapteyn Institute, lecturer (∼50 students) — 09/25–11/25

*Exoplanets*, MSc course, Kapteyn Institute, lecturer (∼20 students) — 10/24–02/25

*Star and Planet Formation*, MSc course, Kapteyn Institute, co-lecturer (∼30 students) — 02/25–04/25

*Exoplanets*, MSc course, Kapteyn Institute, lecturer (∼20 students) — 10/23–02/24

*Hands-on Numerical Astrophysics School for Planet Formation Sciences*, summer school, DFG priority programme 1992 "Exoplanet Diversity", Rauenberg, lecturer (∼30 students) — 08/23

*Star and Planet Formation*, MSc course, Kapteyn Institute, co-lecturer (∼30 students) — 02/23–04/23

*Life and Information*, BSc seminar, University of Groningen, guest lecturer (∼30 students) — 11/22

*Origin and role of volatiles in the early solar system and planets*, block course, University of Bayreuth, lecturer (∼20 students) — 01/20

*Planetary Physics and Chemistry & Geophysics II*, ETH Zurich, substitute lecturer (∼50 students) — 09/17–05/18

*Physics I* as minor subject, ETH Zurich, exercise/exam design (∼100 students) — 09/15–02/16

*Physics I+II* as minor subject, ETH Zurich, teaching assistant (∼30 students) — 02/14–02/15

*Scientific computing*, Göttingen University, teaching assistant (∼20 students) — 09/13–02/14

*Basic physics lab course*, Göttingen University, lab assistant (∼10 students) — 09/13–02/14

*Introduction to Astro- and Geophysics*, Göttingen University, teaching assistant (∼20 students) — 03/13–08/13

*Basic physics lab course*, Göttingen University, lab assistant (∼10 students) — 09/12–02/13

## External/Co-Examination & Defence Committees

*24 examinations in total, incl. 7 PhD, 2 MSc, 15 BSc*

Robin Malaca, "Damping Oscillations in Non-LTE Models", BSc thesis, supervisor: Floris van der Tak, University of Groningen, NL — 07/26

Mariasole De Amicis, "The Atmospheric Composition of WASP-39b: A Broad-Wavelength Band Analysis with JWST", BSc thesis, supervisor: Quentin Changeat, University of Groningen, NL — 07/26

Anastasia Perrotta, "Revisiting the target selection metric for the Ariel telescope", BSc thesis, supervisor: Quentin Changeat, University of Groningen, NL — 07/26

Wout Sijtsma, "Testing the validity of the Shallow Water Equations in Hot Jupiter modeling using the pseudo-spectral PDE solver Dedalus 3", BSc thesis, supervisor: Quentin Changeat, University of Groningen, NL — 07/26

Sacha Veenstra, "Thermal dissociation and atmospheric structure: how do they affect the retrieved atmospheric composition of the ultra-hot Jupiter WASP-18b?", BSc thesis, supervisor: Quentin Changeat, University of Groningen, NL — 07/26

Aditya Arabhavi, "Unraveling the Planet-Forming Materials Around Young Stars with JWST", PhD thesis, supervisor: Inga Kamp, University of Groningen, NL — 09/25

Zalán Novák, "Rotational Excitation of Interstellar OCS: Non-LTE Line Analysis and Temperature-Density Diagnostics Using RADEX", BSc thesis, supervisor: Floris van der Tak, University of Groningen, NL — 07/25

Emma F. van der Vinne, "Atmospheric retrieval of Jupiter, observed as an exoplanet, using the TauREx code", BSc thesis, supervisor: Quentin Changeat, University of Groningen, NL — 07/25

Tristan Popken, "Accelerated chemistry for exoplanet atmospheres via grid interpolation", BSc thesis, supervisor: Quentin Changeat, University of Groningen, NL — 07/25

Fran Štimac, "Unreliable Pixel Correction in JWST Exoplanet Data: Identifying, classifying, and correcting for unreliable pixels on the NIRSpec detector", BSc thesis, supervisor: Quentin Changeat, University of Groningen, NL — 07/25

Jesse Reurink, "A Small Population Study Across the Radius Valley using JWST NIRSpec G395H", MSc thesis, supervisor: Quentin Changeat, University of Groningen, NL — 07/25

Mark Oosterloo, "Tracking CHNOS during the early stages of planet formation with a model-laboratory approach", PhD thesis, supervisors: Inga Kamp, Wim van Westrenen, University of Groningen, NL — 01/25

Francisco Ardévol Martinez, "Machine learning for exoplanet characterisation in the JWST era", PhD thesis, supervisors: Inga Kamp, Paul I. Palmer, Michiel Min, University of Groningen, NL — 10/24

Zara Siyal, "Comparison and characterisation of properties of exoplanets and their host stars", BSc thesis, supervisors: Amina Helmi, Tom Callingham, University of Groningen, NL — 07/24

Tycho Hebels, "The Effects of Dust Grain Surface Reactions on the HDO/$H_2O$ Ice Ratio in Protoplanetary Disks", BSc thesis, supervisor: Inga Kamp, University of Groningen, NL — 07/24

Kevin de Man, "Radiation processing of grains in protoplanetary disks", BSc thesis, supervisor: Inga Kamp, University of Groningen, NL — 07/24

Ioannis Panagiotou, "Evolution of the pressure equilibrium and condensation in idealised simulations", BSc thesis, supervisors: Filippo Fraternali, A. Dutta, University of Groningen, NL — 07/24

Keavin Moore, "Coupled Models of Water on Terrestrial Planets Orbiting M-Dwarfs", PhD thesis, supervisor: Nicolas B. Cowan, McGill University, CA — 12/23

Roos H. Voorhoeve, "Enhancing and expanding the applicability of ReverseRADEX for accurate astrophysical insights", BSc thesis, supervisor: Floris van der Tak, University of Groningen, NL — 12/23

Bayron Armando Portilla Revelo, "Closing the gap between theory and observations of planet-forming disks with radiative transfer models", PhD thesis, supervisors: Inga Kamp, Ewine van Dishoeck, University of Groningen, NL — 12/23

Nynke Visser, "The influence of the ice-albedo feedback on the long term carbon cycle of Earth-like exoplanets", MSc thesis, supervisors: Inga Kamp, Mark Oosterloo, Dennis Höning, University of Groningen, NL — 08/23

Raymon P. Ferdinand, "Outgassing and Exoplanet Atmospheres: A study of the effects of outgassing on the temperature and composition of the secondary atmospheres of rocky exoplanets", BSc thesis, supervisors: Floris van der Tak, Michiel Min, University of Groningen, NL — 06/23

Yapeng Zhang, "Isotopes and the characterization of extrasolar planets", PhD thesis, supervisors: Ignas Snellen, Laura Kreidberg, Leiden Observatory, NL — 06/23

Mantas Zilinskas, "Lava Worlds: Characterising atmospheres of impossible nature", PhD thesis, supervisors: Yamila Miguel, Ignas Snellen, Leiden Observatory, NL — 05/23

## Refereeing & Panel Service

AGU Advances, Astronomy & Astrophysics, Astrophysical Journal, Astrophysical Journal Letters, Earth and Planetary Science Letters, Elements, Icarus, Journal of Geophysical Research: Planets, Monthly Notices of the Royal Astronomical Society, Nature, Nature Astronomy, Nature Communications, Planetary Science Journal, Proceedings of the National Academy of Sciences, Publications of the Astronomical Society of Australia, Science, Science Advances

Belgian Research Foundation (FWO), Cambridge University College Fellowships, Dutch Research Council (NWO), European Research Council (ERC), Max Planck Society (MPG), NASA Emerging Worlds program, NASA Hubble Fellowship Program (NHFP), Natural Sciences and Engineering Research Council of Canada (NSERC), Swiss National Science Foundation (SNSF), UK Research and Innovation (UKRI), US National Science Foundation (NSF)

## National & International Collaborations

INT HARPS3 Semester 26B Program *High-Precision RVs for an Expanded Census of Stellar and Planetary Compositions* (62.2 hr, PI: Lichtenberg), PI — 07/26–

INT HARPS3 Semester 26B Program *HARPS3-NEAR: A Radial-Velocity Census of the Nearest Sun-like Stars* (61.5 hr, PI: Snellen), Co-I 07/26–

JWST Cycle 4 GO Program *Surveying Hellish Worlds: Lava Planets as Time Capsules of Thermal Evolution* (100.5 hr, PI: Dang), Co-I 04/25–

Dutch National Science Foundation (NWO) Consortium PRELIFE – Pathways, Reactions, and Environments leading to LIFE, Co-I 11/24–

JWST Cycle 2 GO Program *Phase Curve Observations of TOI-561 b To Study Atmosphere-Interior Exchange* (44.3 hr, PI: Teske), Co-I 02/23–

ESA PLATO Consortium, member of Exoplanet Science WPs *Atmospheres of PLATO Terrestrial Planets* (WP116400) and *Climate/Atmospheres of PLATO Habitable Zone Planets* (WP116530); Core Science Team member since 2025 12/21–

Co-I (since 2024, previously associate member) of Carnegie-led research program *AEThER: Atmospheric Empirical, Theoretical, and Experimental Research* 09/21–

Associate member of NASA research program Alien Earths: Searching for Habitable Worlds in Nearby Planetary Systems 03/21–

Co-lead of the *Science Team* of the LIFE Collaboration 02/21–01/26

Member of the *Quantifying Habitability* working group of NASA's Nexus for Exoplanet System Science 09/20–

ESA Ariel Consortium, member of the *Exoplanetary Interiors* working group 04/20–

Postdoctoral Fellow of the Simons Collaboration on the Origins of Life (SCOL project) 10/19–09/22

Member of the Large Interferometer for Exoplanets (LIFE) Collaboration 01/19–

## Conference & Workshop Organisation

*28 events in total, incl. 8 as main organiser*

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|---|---|---|
| Rocky Worlds Conference Series (main organiser) | Chair | 10/22– |
| Lorentz Center Workshop *Characterizing Habitable Exoplanets With Interdisciplinary Expertise*, Leiden, NL | SOC | 04/26 |
| Rocky Worlds IV (main organiser), Groningen, NL | Chair | 01/26 |
| *Science and Technology for the Era of LIFE*, Barcelona, ES | SOC | 11/25 |
| Lorentz Center Workshop *Atmospheric and interior evolution of planetary magma oceans* (main organiser), Leiden, NL | Chair | 10/25 |
| *Exoclimes VII*, Montréal, CA | SOC | 07/25 |
| *Exoplanets V*, Leiden, NL | Convenor | 06/24 |
| Rocky Worlds III, ETH Zurich, CH | SOC | 01/24 |
| Annual LIFE Science Team Workshop, Berlin, DE | Co-Chair | 09/23 |
| Goldschmidt 2023, *Unraveling planetary interiors with extraterrestrial samples and space data*, Lyon, FR | Convenor | 07/23 |
| *Origins of Solar Systems* Gordon Research Seminar (main organiser), South Hadley, US | Co-Chair | 06/23 |
| *The (geo)chemistry of terrestrial planet formation*, Groningen, NL | SOC & LOC | 03/23 |
| Rocky Worlds Discussions, monthly virtual seminar series (main organiser) | SOC | 10/22–09/24 |
| Rocky Worlds II (main organiser), University of Oxford, UK | Chair | 07/22 |
| AGU 2021 session *Multidisciplinary perspectives on the formation and early evolution of terrestrial worlds*, New Orleans, US | Convenor | 12/21 |
| ESA PLATO Exoplanet Atmospheres Workshop, Berlin, DE | SOC | 12/21 |
| Lorentz Center Workshop *LIFE – Large Interferometer for Exoplanets* (main organiser), Leiden, NL | SOC | 06/21 |
| AGU 2020 session *Accretion and differentiation of rocky planets: perspectives from geophysics, geochemistry, & astronomy*, San Francisco, US | Convenor | 12/20 |
| *LIFE Workshop III*, Liège, BE | SOC | 12/20 |
| *Threats from the surroundings*, ESO, Garching, DE | SOC | 11/20 |
| *Exoplanets III*, Heidelberg, DE | Moderator | 07/20 |
| *LIFE – Large Interferometer for Exoplanets* (main organiser), virtual, ETH Zurich | SOC | 05/20 |
| *3rd Oxford Meeting on Planets* (main organiser), University of Oxford, UK | SOC | 12/19 |
| *Exoclimes V*, University of Oxford, UK | LOC | 08/19 |

*2nd Oxford Meeting on Planets* (main organiser), University of Oxford, UK — Chair — 12/18

*Water during planet formation and evolution*, University of Zurich, CH — SOC — 02/18

*The Origins of Volatiles in Habitable Planets*, University of Michigan, US — LOC — 10/17

*Water during planet formation and evolution*, PlanetS internal workshop, University of Zurich, CH — SOC — 11/16

## Advanced Teaching & Research Training Courses

### Teaching, Education & Supervision

*Academic Leadership*, University of Groningen, NL — 09/25

*University Teaching Qualification* (UTQ), University of Groningen, NL — 02/24

*Introduction for new PhD supervisors*, University of Groningen, NL — 10/23

*Active Bystander Training*, Kapteyn Institute, NL — 04/23

*The importance of the first PhD year*, University of Groningen, NL — 03/23

*Coaching PhD students*, University of Groningen, NL — 01/23

*Inclusive Leadership: The Power of Workplace Diversity*, University of Colorado, US (online) — 06/21

*University Teaching*, University of Hong Kong, HK (online) — 05/21

*Researcher Management and Leadership Training*, University of Colorado, US (online) — 05/21

*Learning to Teach*, ETH Zurich, CH — 12/15

### Research Schools

Winter school *Chronology of the Formation of the Solar System VI*, Les Houches, FR — 02/17

Winter school *Planetary Geodynamics*, TRR 170 Late Accretion onto Terrestrial Planets, Hodenhagen, DE — 11/16

Winter school *From Protoplanetary Disks to Planet Formation*, 45$^{th}$ Saas-Fee Advanced Course, Les Diablerets, CH — 03/15

Summer school *ISSAC 2013: Star & Planet Formation*, UC Santa Cruz, US — 08/13

## Outreach & Public Engagement

*50+ outreach activities, incl. 14 media interviews & press features, 4 public talks & 2 panels, 16 blog posts & videos, 20 peer-reviewed Astrobites articles*

### Media Interviews & Press

Media interview, *Astronomen ontdekken de eerste rotsachtige exoplaneet met een atmosfeer*, KIJK Magazine — 07/26

Press coverage for Nicholls et al. Nature Astronomy 2026 on the molten sulfur world L 98-59 d, primary feature *Molten Sulfurous World Blurs Exoplanet Categories* (Sky & Telescope), also covered in ABC News, Space.com, University of Groningen news, and others — 03/26

Media interview, *Stormy planets and an unexpected atmosphere*, FSE Science Newsroom, University of Groningen — 12/25

Media interviews for NASA, Space.com, phys.org, ScienceDaily, and others on TOI-561 b atmosphere detection — 11–12/25

Interview, *Quickfire Questions with Tim Lichtenberg*, Europlanet Magazine — 09/25

Media interview, *This Blazing Exoplanet Breaks All the Rules about Alien Atmospheres*, Scientific American — 09/25

Media interview, *Exoplanet plate tectonics: A new frontier in the hunt for alien life*, New Scientist — 12/24

Media interview, podcast *Die knifflige Suche nach außerirdischem Leben*, Deutschlandfunk — 12/24

Media interview, *What is a presumed sign of life doing on a dead comet?*, Science News — 04/24

Biographical interview, *Welt der Physik*, Year of Science 2023, German Ministry for Science and Education — 04/23

Media interview, *Planeet en ster groeien samen op*, New Scientist Netherlands — 12/22

Press coverage for Bonsor+2023 *Nature Astronomy*, University of Groningen, ScienceDaily, and others — 11/22

Press coverage for Lichtenberg+2021 *Science*, phys.org, ScienceDaily, UPI, SciTechDaily, and 36+ outlets — 01/21

Press coverage for Lichtenberg+2019 *Nature Astronomy*, ETH Zurich, NCCR PlanetS, and 21+ outlets — 02/19

## Public Talks & Panels

Public talk *Formation and Evolution of Terrestrial Worlds*, DOT Planetarium Groningen 05/24

Public talk *Entstehung und Entwicklung terrestrischer Welten* (∼100 attendees), Max Planck Institute for Solar System Research, Göttingen, DE 05/24

Panel member, Golden Webinars in Astrophysics: Lynnae Quick, *Cryovolcanism on Ocean Worlds* 10/21

Panel member, Golden Webinars in Astrophysics: Frances Westall, *Biosignatures and the Search for Extraterrestrial Life* 10/21

Public talk *Wasser, Wüste, oder Lava: wie entstehen erdähnliche Planeten?*, Planetarium Göttingen, published on Urknall, Weltall und das Leben (>34'500 views) 11/20

Public talk *Distant worlds in the computer: planet formation via gravitational instabilities*, Göttingen University, DE 03/14

## Blog Posts & Videos

Blog articles and paper summary videos published on Medium/Forming Worlds, YouTube, and NCCR PlanetS.

Hidden water in magma ocean exoplanets (video) 10/21

The turbulent hearts of shrouded super-Earths (blog + video) 05/21

Early carbon depletion of terrestrial worlds from the outside-in (blog + video) 05/21

Molten exoplanets as a window into the earliest Earth (blog + video) 01/21

Formation of the Solar System in two steps (blog + video) 01/21

Entstehung des Sonnensystems in zwei Phasen, popular article in *Sterne und Weltraum* 08/21

Two emergent faces of rocky planetary systems (blog) 02/19

Planetesimal magmatism as a key to the chemical assembly of terrestrial planets (blog) 01/19

Spotting protoplanet collision afterglows in the stellar neighbourhood (blog) 11/18

Impact splash chondrule formation during planetesimal recycling (blog) 11/17

Supernova ejecta polluting young planetary systems (blog) 08/16

On the variety of asteroid interiors (blog) 04/16

Author and referee for Astrobites, a graduate student-run blog introducing undergraduates to astrophysical research (20 peer-reviewed articles, >13'000 views; 24 reviews; application committee 2015, 2016) 09/14–08/19

Why open research practices are a clever move 12/16

Modest chaos in the early solar system 11/16

Driving planetary dynamos by giant impacts 08/16

Water worlds – self-arrests, thermostats and long-term climate stability 07/16

Massive circumstellar disks accrete faster than low-mass ones 05/16

Grand fireworks from the Local Super-Bubble 04/16

Bridging the gap: asteroid collisions without quantum foam 03/16

The tempestuous adolescence of circumstellar disks 02/16

Sustainable climates on greenhouse super-Earths? 11/15

Ripping apart asteroids to account for Earth's strangeness 11/15

Dry out super-Earths via giant impacts 10/15

Triggered fragmentation in self-gravitating disks 09/15

Giant planets from far out there 08/15

Nature's Starships Vol. II – A Hitchhiker's Guide to the Bloody Cold Beginnings 07/15

Nature's Starships Vol. I – Ride in on Shooting Stars 06/15

Circumplanetary disks… soon! 05/15

Hot Shots: How to trigger star formation in the early Universe by supernova blast-waves 05/15

Why is star formation so inefficient? 04/15

Are extrasolar worlds more likely to be water-rich? 03/15

Sneak a peek on the cause of death of protoplanetary disks 02/15

### Other Activities

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| Letters to a Pre-Scientist pen pal | 2019–2020 |
| *Expedition Solar System*, Focus Terra science exhibition, ETH Zurich | 03/18 |
| Figures and media for popular science article *Exotische Welten* in *Sterne und Weltraum* | 08/15 |
| Zurich Scientifica 2015, *Licht und Astronomie*, public science exhibition, ETH Zurich | 09/15 |
| Astrobites reddit Ask-Me-Anything | 06/15 |
| Q&A visit, primary school, Zurich Seefeld | 06/15 |
| 3D rendering video for Bovino et al. (2014), ApJL 790, L35 (phys.org, INAF, HPC Wire) | 07/14 |

## Academic Associations

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| Royal Netherlands Astronomical Society | 05/24– |
| European Association of Geochemistry | 07/21– |
| International Astronomical Union | 05/21– |
| American Geophysical Union | 01/20– |
| European Astronomical Society | 06/19– |
| Royal Astronomical Society | 06/19– |
| Europlanet Society | 03/19– |
| European Geosciences Union | 01/19– |
| Swiss Society for Astrophysics and Astronomy | 01/15–01/24 |
| German Astronomical Society | 11/13– |
| German Physical Society | 11/10– |